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BRMK557

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Research Methodology and IPR

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Identify the meaning of Research and brief out the objective and motivation in engineering research.	10	L1	CO1
	b.	Explain brief about research cycle and verify with the research flow diagram.	10	L1	CO1
OR					
Q.2	a.	Identify the types of engineering research and briefly explain them.	10	L1	CO1
	b.	Explain about the different types of research misconduct.	10	L1	CO1
Module – 2					
Q.3	a.	Explain about the importance of literature review and technical reading.	10	L2	CO2
	b.	Mention the various benefits of bibliographic databases.	10	L1	CO2
OR					
Q.4	a.	Identify the impact of technical reaction and brief about it.	10	L1	CO2
	b.	Enumerate the impact of title and keywords on citation with example.	10	L2	CO2
Module – 3					
Q.5	a.	Define Intellectual properties and explain about its types.	10	L1	CO3
	b.	Explain about the key aspect of patent law.	10	L2	CO3
OR					
Q.6	a.	Explain about the assessment of novelty.	10	L1	CO3
	b.	Brief about the patent procedure in India.	10	L1	CO4
Module – 4					
Q.7	a.	Mention and brief about the justification for copyright law.	10	L2	CO4
	b.	Explain about the basic concepts of under lying copyright law.	10	L1	CO4
OR					
Q.8	a.	Brief about the various representations of sound recordings.	10	L2	CO5
	b.	Explain about TRIPS agreement in detail.	10	L1	CO5

Module – 5					
Q.9	a.	Explain about the justification of protection designs.	10	L2	CO5
	b.	Brief about the excluded subjected matter in the context of design protection.	10	L1	CO5
OR					
Q.10	a.	What are the rights of the owner of designs? Explain.	10	L1	CO5
	b.	Brief about the Assignment of Design Rights.	10	L1	CO5

CBCS SCHEME - Make-Up Exam

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BRMK557

Fifth Semester B.E/B.Tech. Degree Examination, June/July 2025 Research Methodology and IPR

Time: 3 hrs.

Max. Marks:100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
1	a.	Define Engineering Research. Explain the objectives of Engineering Research.	10	L1	CO1
	b.	Explain the types of Engineering Research.	10	L2	CO1
OR					
2	a.	Explain the ethical issues related to authorship.	10	L2	CO1
	b.	Explain the motivation of research in engineering.	10	L2	CO1
Module – 2					
3	a.	Explain the steps followed for literature review.	10	L2	CO2
	b.	Explain Web of science and Google Scholar.	10	L2	CO2
OR					
4	a.	Differentiate between citations and attributions.	10	L2	CO2
	b.	Explain the impact of Title and Keywords on citations.	10	L3	CO2
Module – 3					
5	a.	List the major amendments in IP laws and acts in India.	10	L3	CO3
	b.	Explain the types of Patents Applications.	10	L1	CO3
OR					
6	a.	Explain the role of IP in economic and cultural development of the society.	10	L2	CO3
	b.	Explain the different parts of a patent application with emphasis on importance of claims.	10	L2	CO3
Module – 4					
7	a.	Illustrate the steps involved in the process of trademark registration using flow chart.	10	L1	CO4
	b.	Explain the process of trademark registration.	10	L2	CO4

OR					
8	a.	Explain the types of copyright infringement with examples.	10	L2	CO4
	b.	Explain the key eligibility criteria that a trademark must need to qualify for trademark protection.	10	L2	CO4
Module – 5					
9	a.	Summarize the non protectable industrial designs in India.	10	L3	CO5
	b.	Discuss the case study of Basmati patent.	10	L3	CO5
OR					
10	a.	Define Geographical Indications (GI) with an example. What are the rights granted to GI holders?	10	L2	CO5
	b.	Using the concept of patent discuss the case study of Neem Patent.	10	L3	CO5

CBCS SCHEME

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BRMK557

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026 Research Methodology and IPR

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define the term Research. Explain the research flow cycle with the help of a neat diagram.	07	L2	CO1
	b.	Interpret the factors that motivate Engineering Research.	07	L2	CO1
	c.	With the help of neat diagram, recall the 3 broad categories of developing and accessing knowledge in research.	06	L1	CO1
OR					
Q.2	a.	What do you mean by Ethics? Recall the importance of practicing ethics in engineering research.	07	L1	CO1
	b.	Write short notes on the following : i) Fabrication ii) Plagiarism	08	L2	CO1
	c.	Explain in brief the 3 ways to credit the Research Contributions.	05	L2	CO1
Module – 2					
Q.3	a.	Explain briefly the goals of conducting literature review in academic research.	10	L2	CO2
	b.	Explain how existing knowledge will act as a foundation for new knowledge.	10	L2	CO2
OR					
Q.4	a.	Define Citation. Mention and explain the types of citations which fail to achieve their goals in benefiting the reader.	10	L2	CO2
	b.	With the help of neat diagram, illustrate how knowledge flows through a citation network.	10	L2	CO2
Module – 3					
Q.5	a.	What is Intellectual Property? Discuss the role of IP in economic and cultural development of the society.	10	L2	CO1
	b.	List the type of inventions which are eligible for patenting and which are not patentable?	10	L1	CO3

OR

Q.6	a.	Define the term Patent. List and explain the conditions that must be met for obtaining a patent protection.	10	L2	CO3
	b.	With the help of neat diagram, explain the process of patenting an invention.	10	L2	CO3

Module – 4

Q.7	a.	What is Copy Right? Explain the criteria that a work must meet to qualify for Copy Right Protection.	10	L2	CO4
	b.	With the help of flowchart, explain the important steps involved in the process of copy right registration.	10	L2	CO4

OR

Q.8	a.	What is a Trademark? Explain eligibility criterias for trademark and designation of Trademark Symbols.	10	L2	CO4
	b.	With the help of neat flowchart, explain the process of trademark registration.	10	L2	CO4

Module – 5

Q.9	a.	What is an Industrial Design? Explain the procedure for registering an industrial design.	10	L2	CO5
	b.	Explain the classification of Industrial Design.	10	L2	CO5

OR

Q.10	a.	What is GI? Explain the mechanism available for the protection of rights related to GI.	10	L2	CO5
	b.	Discuss the case study of curcuma (turmeric) patent.	10	L2	CO5

Model Question Paper-1 with effect from 2022(CBCS Scheme)

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Fifth Semester B. Tech Degree Examination
Research Methodology and Intellectual Property Rights

TIME: 03 Hours**Max.Marks:100**

Note: Answer any FIVE full questions, choosing at least ONE question from each MODULE

Module-1			*Bloom's Taxonomy Level	COs	Marks
Q.01	a	List the different types of research misconduct and provide a brief explanation for each one.	L1	1	10
	b	Define Engineering Research and list its aims and objectives	L2	1	10
OR					
Q.02	a	What ethical considerations and responsibilities should be taken into account when determining authorship in engineering research?	L2	1	10
	b	Enlist and Explain different types of engineering research	L2	1	10
Module-2					
Q.03	a	How does the new and existing knowledge can contribute to the research process? Explain with relevant points.	L2	2	10
	b	How can researchers effectively use search engines to find relevant literatures in their field?	L2	2	10
OR					
Q.04	a	Explain how knowledge flows through a citation network using a flow diagram.	L2	2	10
	b	What is impact of Title and Keywords on citations? Explain Citation based Knowledge flow	L1	2	10
Module-3					
Q.05	a	Describe Intellectual Property Rights and List its types?	L2	3	10
	b	Explain the following major steps involved in the process of Patent Registration: i)Prior Art Search ii)Choice of application to be filed iii)Pre-grant opposition	L2	3	10
OR					
Q.06	a	In which circumstances Indian residents are not required to file a patent application first in India to get patent protection in another country? Explain.	L2	3	10
	b	Explain step by step process of obtaining the patent from the initial idea to the grant of patent.	L2	3	10
Module-4					
Q.07	a	Using a flowchart, explain the steps involved in the process of Trademarks Registration.	L2	4	10
	b	Define the term Copyright and write its classes. What are the two exclusive rights owned by the copyright owner? Explain briefly.	L1	4	10
OR					
Q.08	a	Using a flowchart, explain the steps involved in the process of Copyright Registration.	L2	4	10
	b	Copyright Infringements, Copyright Infringement is a Criminal Offence. Copyright Infringement is a Cognizable Offence Explain.	L2	4	10

Module-5					
Q.09	a	Explain the process of industrial design registration.	L2	5	10
	b	Define Geographical Indications (GI) with an example. What are the rights granted to GI holders?	L2	5	10
OR					
Q.10	a	Explain Case study of Curcuma (Turmeric) Patent, Case study of Neem Patent.	L2	5	10
	b	Using a flow chart, explain the process of GI registration	L1	5	10

*Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the Cos and Pos to be attained by every bit of questions.

Model Question Paper-2 with effect from 2022(CBCS Scheme)

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Fifth Semester B. Tech Degree Examination Research Methodology and Intellectual Property Rights

TIME: 03 Hours

Max.Marks:100

Note: Answer any FIVE full questions, choosing at least ONE question from each MODULE

Module-1			*Bloom's Taxonomy Level	COs	Marks
Q.01	a	What are the key ethical issues related to authorship? Explain each one.	L1	1	10
	b	What are the three broad categories of developing and accessing Knowledge in research? Explain with a diagram.	L2	1	10
OR					
Q.02	a	Explain Fabrication, Falsification and Plagiarism related to engineering Research	L2	1	10
	b	Discuss the different types of engineering research. Clearly point out the Differences between all of them with examples.	L2	1	10
Module-2					
Q.03	a	Explain the various steps involved in the critical and creative reading process.	L2	2	10
	b	What are the key features of the bibliographic database of the Web of Science (WoS), and how is it commonly used in research?	L2	2	10
OR					
Q.04	a	What types of citations fail to achieve their goal and do not benefit the reader? Explain.	L2	2	10
	b	Explain the most common styles for citation used by engineers during research, and provide an example.	L1	2	10
Module-3					
Q.05	a	Discuss the Design registration procedure of patent by using a flowchart.	L2	3	10
	b	Discuss the history of Intellectual Property Rights in India	L2	3	10
OR					
Q.06	a	What strategies are involved in the commercialization of a patent?	L2	3	10
	b	What are utility models, and how do they differ from patents?	L2	3	10
Module-4					
Q.07	a	Explain the process of copyright registration? What are the benefits for the copyright holders?	L2	4	10
	b	Explain by using flowchart, steps involved in trademark registration?	L1	4	10
OR					
Q.08	a	Explain the criteria that an original work must meet to qualify for copy right protection?	L2	4	10
	b	What are the roles and functions of the copyright board and the copyright Society in administering copyright laws and regulations?	L2	4	10

Module-5					
Q.09	a	Explain the classification of Industrial Designs and design registration trends in India	L2	5	10
	b	Explain the famous case law between Apple Inc Vs Samsung Electronics Co. related with Industrial Design Rights.	L2	5	10
OR					
Q.10	a	How would you describe the overall ecosystem and significance of geographical indications in India?	L2	5	10
	b	Using a flowchart, Explain the process of GI registration.	L1	5	10

*Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the Cos and Pos to be attained by every bit of questions.